Freelance Electronics Engineer EasyEDA Redesign and Prototype Validation

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Project overview

I am looking for an electronics engineer to redesign and industrialize an existing ESP32-based controller PCB. The current PCB is functional but unreliable, with intermittent Ethernet startup initialisation failures, sensitivity to nearby conductive objects, and suspected ground-return, power, reset, and SPI signal-integrity issues.

Product description

The product is a compact, battery-powered footswitch controller based on an ESP32 module. It includes:

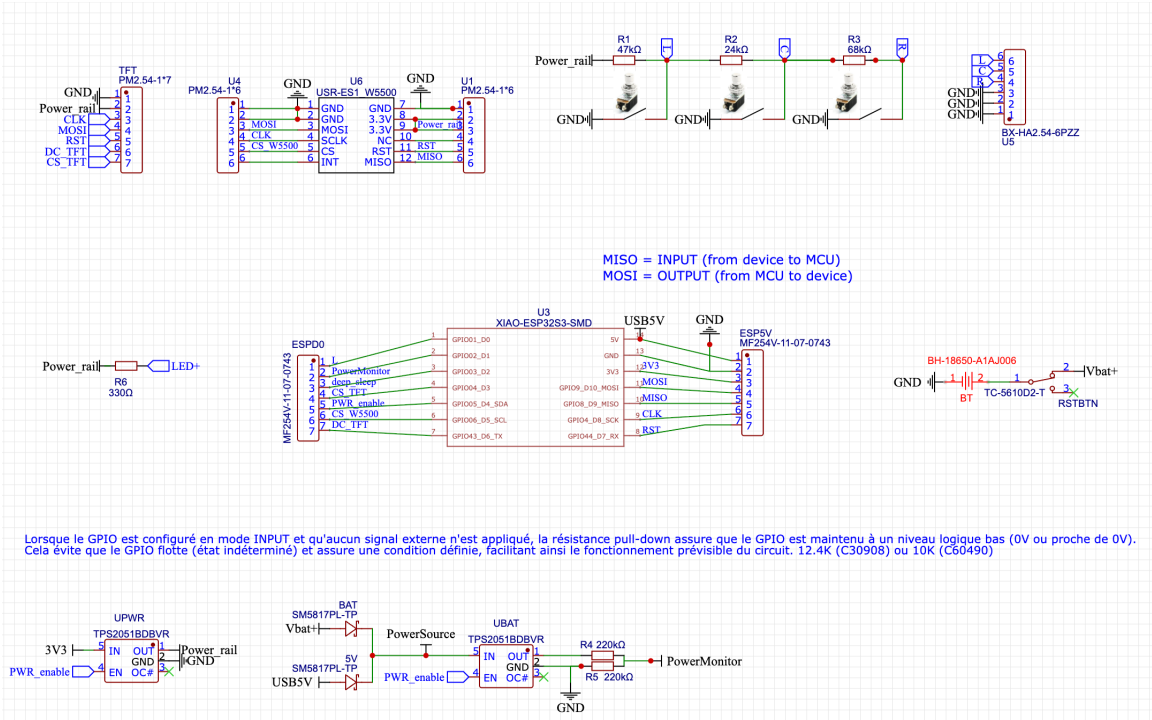
- Three analog footswitch/button inputs using resistor-ladder detection.
- One dedicated digital push button for power-on, shutdown, and startup control.
- A TFT display for user feedback and configuration.
- A W5500 Ethernet interface for wired network connectivity.
- USB connectivity for power, debugging, firmware updates, and configuration.
- Battery-powered operation with a controlled peripheral power rail.
- A shared SPI bus between the TFT display and the W5500 Ethernet controller.

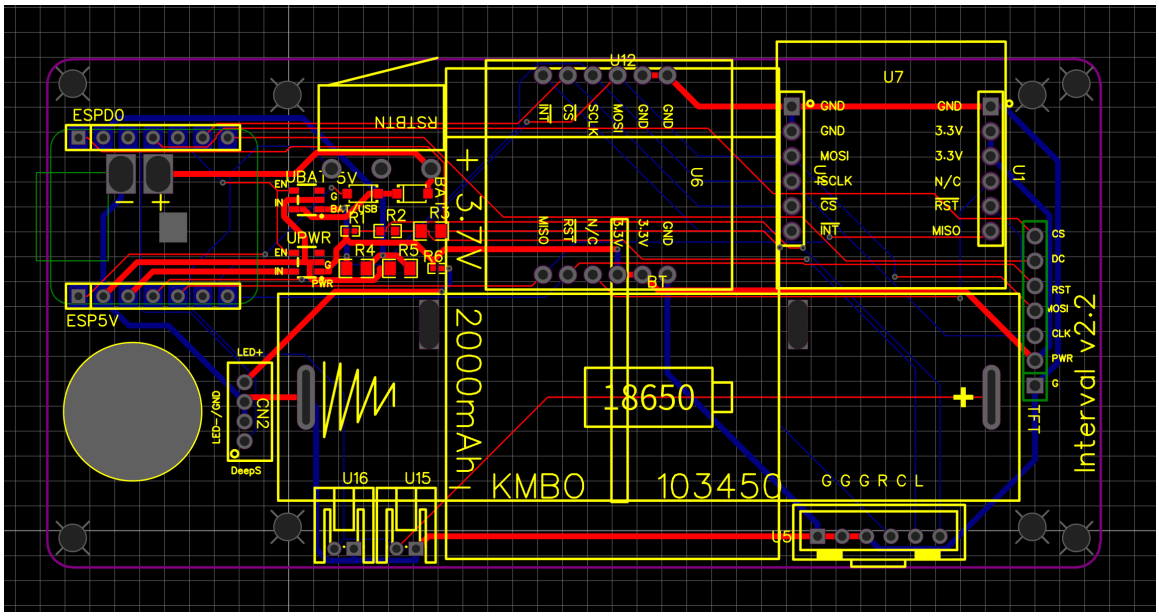
The design must operate reliably from both USB power and 1 18650 battery power and ideally keep the 2 conenctor for 2 2000Mah lithium battery

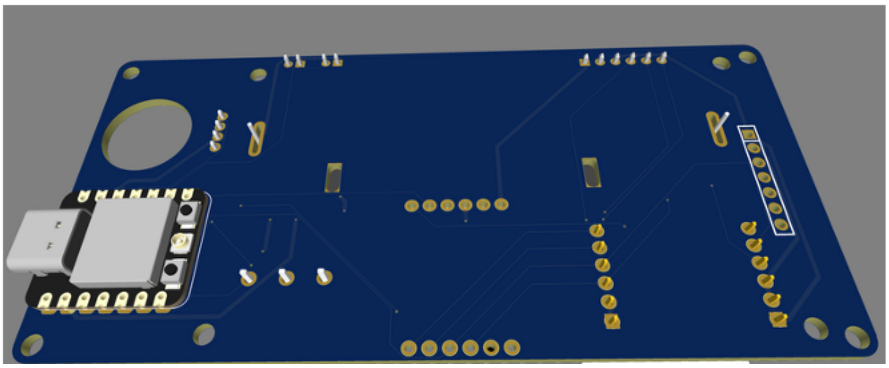
Work must be done on collaborative existing shared easyeda project (online)

Current schematic and reference files

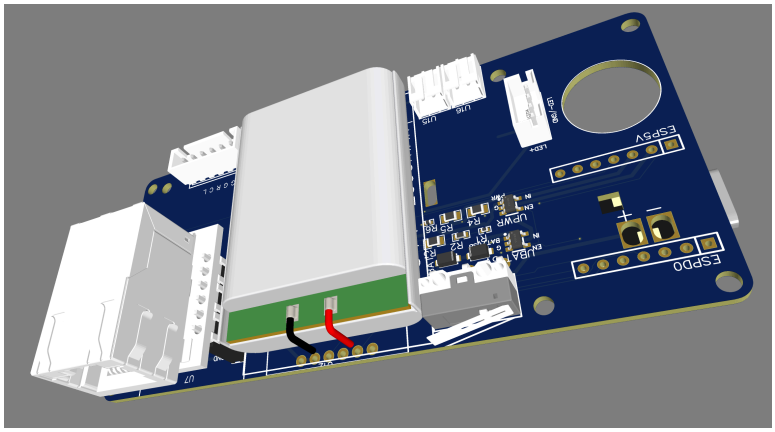
Current schematic: (see easyeda project)


TFT
PM2.54-1*7
GND
Power_rail
CLK
MOSI
RST
DC_TFT
CS_TFT
U4
PM2.54-1*6
U6
USR-ES1_W5500
GND
GND
3.3V
MOSI
SCLK
CS
INT
NC
RST
MISO
U1
PM2.54-1*6
R1
47kΩ
R2
24kΩ
R3
68kΩ
BX-HA2.54-6PZZ
U5
MISO = INPUT (from device to MCU)
MOSI = OUTPUT (from MCU to device)
R6
330Ω
LED+
ESPD0
MF254V-11-07-0743
PowerMonitor
deep_sleep
CS_TFT
PWR_enable
CS_W5500
DC_TFT
U3
XIAO-ESP32S3-SMD
USB5V
ESP5V
MF254V-11-07-0743
BH-18650-A1AJ006
BT
TC-5610D2-T
RSTBTN
Lorsque le GPIO est configuré en mode INPUT et qu'aucun signal externe n'est appliqué, la résistance pull-down assure que le GPIO est maintenu à un niveau logique bas (0V ou proche de 0V).
Cela évite que le GPIO flotte (état indéterminé) et assure une condition définie, facilitant ainsi le fonctionnement prévisible du circuit. 12.4K (C30908) ou 10K (C60490)
UPWR
TPS2051BDBVR
3V3
PWR_enable
Power_rail
GND
BAT
SM5817PL-TP
Vbat+
5V
USB5V
PowerSource
UBAT
R4 220kΩ
R5 220kΩ
PowerMonitor

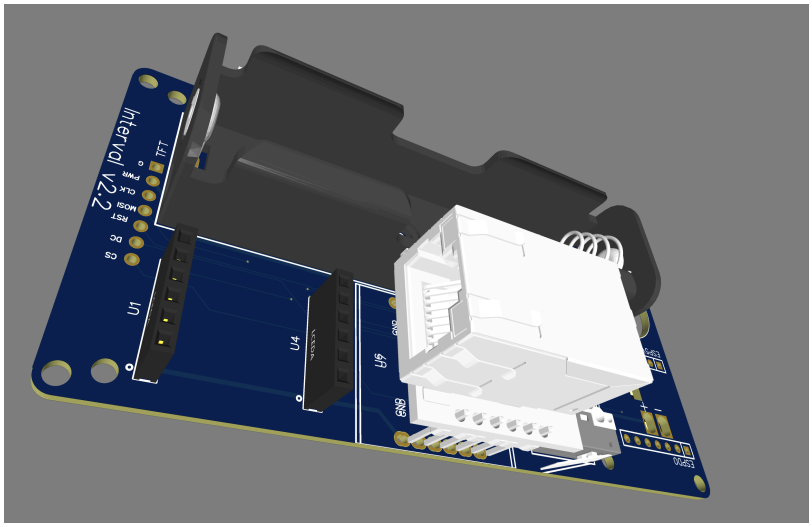

ESPD0
RSTBTN
UBAT
UPWR
ESP5V
LED+
CN2
DeepS
U12
U6
U7
U1
U16
U15
U5
+3.7V 2000mAh
18650
KMBO
103450
Interval v2.2
TFT
BT



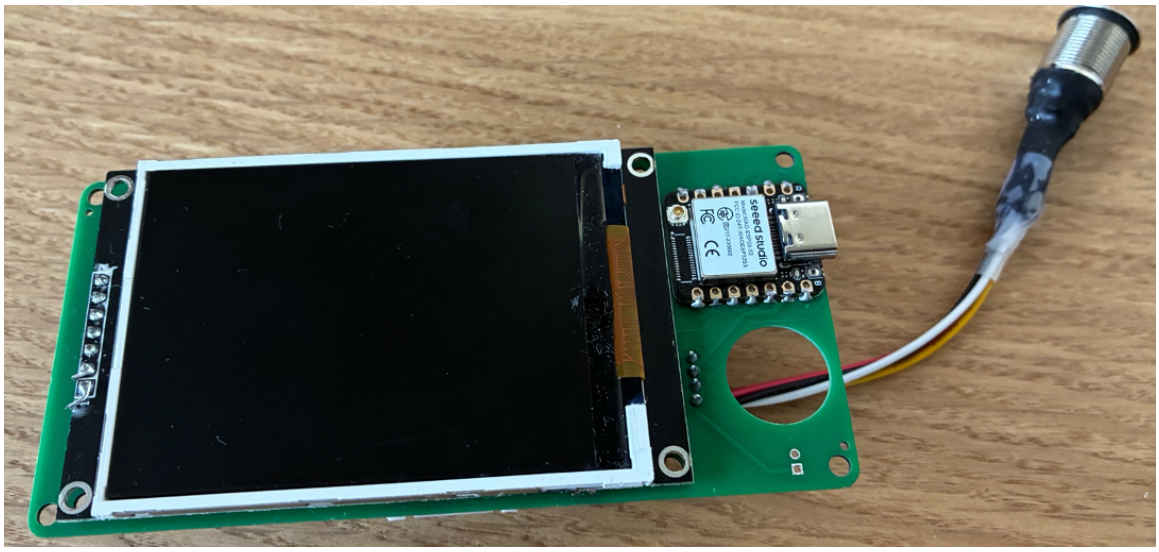
Config 1 : with lithium battery and w5500 in position 1

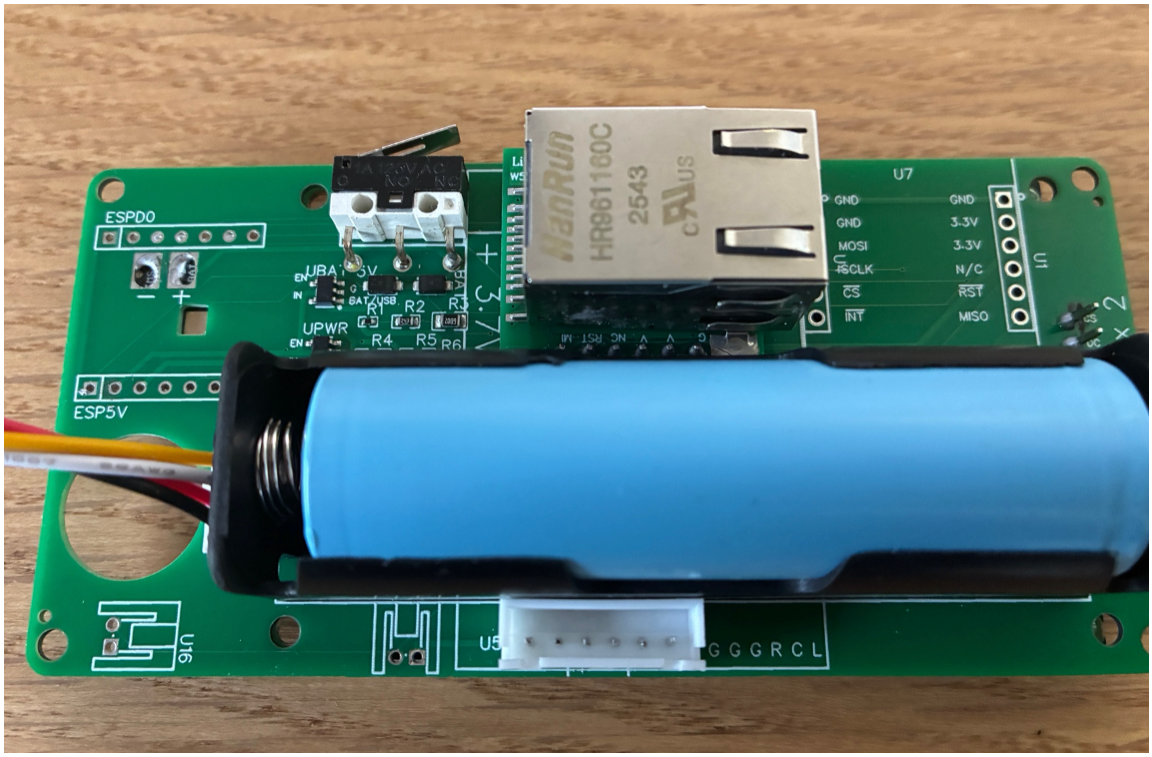


Config 2 : with 18650 battery and w5500 in position 2



Picture of current proto :





Scope of work

1. Recreate and improve the complete design in EasyEDA using the curent design as baseline

2. Work collaboratively in EasyEDA using shared project access and regular design reviews.

3. Use only standard components available from JLCPCB/LCSC so the PCB and components can be ordered directly through JLCPCB.

4. Review and improve the ground-plane strategy, power distribution, decoupling, return-current paths, reset and enable signals, SPI routing, signal integrity, floating inputs, capacitive coupling, protection, startup sequencing, manufacturability, and assembly reliability.

5. Produce a robust prototype suitable for direct JLCPCB assembly.

Mechanical constraints

The following must remain compatible with the existing product and enclosure:

- PCB outline and dimensions.
- Mounting holes and hole locations.
- Battery position and dimensions. (2 possible configs)
- USB connector position.
- ESP32 module position.
- W5500 module/socket position. (2 possible configs)
- TFT connector and other major connector locations.
- Required clearances and enclosure interfaces.

Component cost target

The total component cost must not exceed USD 10 per assembled PCB, excluding PCB fabrication and shipping. Any proposed component that threatens this target must be identified before approval.

Required deliverables

- Complete EasyEDA schematic.
- Complete PCB layout.
- Production-ready Gerber files.
- JLCPCB/LCSC-compatible BOM.
- Pick-and-place/CPL files where applicable.
- Design notes explaining grounding, power, reset, and SPI improvements.
- ERC/DRC-clean project.
- Prototype ordered through JLCPCB.
- Bring-up and validation report.
- Recommendations for any remaining hardware improvements.

Acceptance criteria

The prototype must boot reliably from USB and battery power, start without requiring a connected serial terminal, initialize the ESP32, TFT, and W5500 reliably, maintain stable 3.3 V and ground references, pass repeated cold-start and power-cycle tests, preserve mechanical compatibility, use only JLCPCB/LCSC-sourced components, and remain within the USD 10 component-cost target.

Payment terms

A reasonable fee will be paid for the engineering work performed, including design and documented effort. Final payment will be released only after the EasyEDA design has been used to order and assemble a prototype that meets the agreed acceptance criteria.

Preferred milestones:

- Initial design and schematic milestone Question/Answers session between the 2 of us
- PCB/layout review milestone.
- Prototype-order milestone.
- Final validation and acceptance milestone.

Required experience

Please provide examples of previous work involving ESP32 or similar microcontrollers, W5500 Ethernet, SPI displays, battery-powered products, mixed-signal or compact embedded PCBs, EasyEDA, JLCPCB production, grounding, and signal-integrity debugging.

Please include your estimated cost, schedule, and proposed workflow.